respectively suffer one dimensional elastic collision. Find their velocities after collision.

Solution:

Mass $m_1 = 60$ kg

Mass $m_2 = 30$ kg

$$V_1 = 40\ cms^{-1}$$

$$V_2 = 30\ cms^{-1}$$

Solution:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2$$

$$v_2 = \frac{(m_2 - m_1)}{m_1 + m_2}u_2 + \frac{2m_1}{m_1 + m_2}u_1$$

Substituting the values, we get,

$$v_1 = \frac{(60-30)}{90} \times 40 + \frac{2 \times 30}{90} \times 30$$

$$v_1 = \frac{1}{90}\ [1200+1800]$$

$$= \frac{3000}{90} = 33.3 cms^{-1}$$

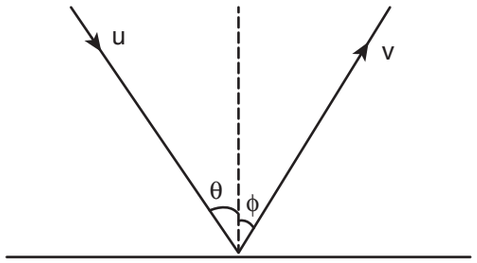
Likewise,

$$v_2 = \frac{(30-60)}{90} \times 30 + \frac{2 \times 60}{90} \times 40$$

$$v_2 = \frac{1}{90}[-900 + 4800]$$

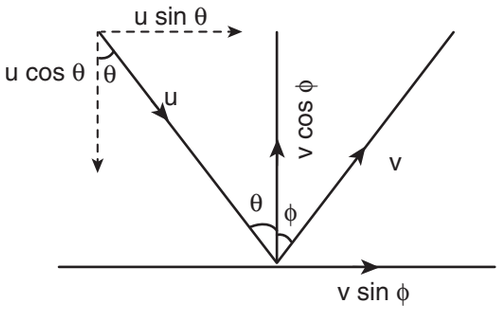
$$= \frac{3900}{90} = 43.3 cms^{-1}$$

6. A particle strikes a horizontal frictionless floor with a speed u at an angle θ with the vertical and rebounds with the speed v at an angle ϕ with the vertical. The coefficient of restitution between the particle and floor is e. What is the magnitude of v?



Solution:

Applying component of velocities,



The x - component of velocity is

$$u\sin\theta = v\sin\phi \qquad (1)$$

The magnitude of y – component of velocity is not same, therefore, using coefficient of restitution,

$$e = \frac{v\cos\varphi}{u\cos\theta} \qquad (2)$$

Squaring (1) and (2) and adding we get

$$v^2 \sin^2\varphi = u^2 \sin^2\theta$$

$$v^2 \cos^2\varphi = e^2u^2 \cos^2\theta$$

adding

$$v^2 = u^2 \sin^2\theta + e^2u^2 \cos^2\theta$$

$$\therefore v^2 = u^2\left[\sin^2\theta + e^2\cos^2\theta\right]$$

$$v = u\sqrt{\sin^2\theta + e^2\cos^2\theta}$$

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